Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination: A Diagnostic Analysis

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Abstract

Temporal abstraction—imagining at variable time scales—is a natural capability of learned world models, yet prior attempts to exploit it in DreamerV3 consistently degrade performance. We present a systematic factorial diagnosis identifying the root cause: a *trained dt-embedding* that conditions the critic on temporal-skip information triggers **Conditioning-Induced Value Overestimation (CIVO)**, a positive-feedback loop in which the critic exploits precise but inaccurate dt-conditioned predictions from the RSSM. Using a normalized value-to-eval (NVE) diagnostic, we show that a 1.5× threshold at 35K+ steps perfectly separates pathological from healthy runs (16/16 accuracy across 4 environments: Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop). Gradient decomposition reveals that 93% of the dt-embedding gradient originates from observation decoders, not the reward head, pinpointing the architectural vulnerability. Restricting imagination to dt=1 eliminates the primary CIVO pathway but exposes a secondary eval-driven instability (Pathway B, 1/4 seeds), demonstrating that surface-level mitigations are insufficient. Notably, TAWM (Nhu et al., 2025) succeeds with dt-conditioning under MPC, confirming that the vulnerability is specific to imagination-based actor-critic architectures. Our findings provide actionable guidance: do not train dt-embeddings for temporal conditioning in imagination-based actor-critic systems without safeguards against the observation-decoder gradient pathway.

1 Introduction

Temporal abstraction—the ability to reason and plan at multiple time scales—has long been a goal of reinforcement learning research, from the options framework [Sutton et al., 1999] to macro-actions and action repetition [Lakshminarayanan et al., 2017]. World-model reinforcement learning, exemplified by the Dreamer family [Hafner et al., 2020, 2023], offers a particularly natural testbed: one can modify the imagination horizon's temporal granularity without changing environment interaction, enabling variable-rate "mental simulation."

Despite this appeal, our systematic exploration of variable-rate imagination in DreamerV3 reveals a striking negative result: *every* configuration that introduces dt>1 imagination degrades asymptotic performance relative to the dt=1 baseline. Fixed dt=2 imagination peaks at 485 (51% of baseline 950) before collapsing; dt=4 reaches only 177 (19% of baseline); learned gating peaks at 541 then collapses to 99.

This paper asks: *why does temporal abstraction fail in world-model RL?*

Through a controlled factorial experiment design, we identify the root cause as **Conditioning-Induced Value Overestimation (CIVO)**: when the dt-embedding is *trained* (receives gradient from the world model's observation decoders), it enables the critic to make precise but systematically biased dt-specific value predictions. The RSSM's prediction fidelity degrades at dt>1 (reward gap 2.78× for trained dt-embedding vs. 1.01× without), but the trained embedding creates the illusion of reliable conditioning, triggering a positive-feedback loop: overestimated values → actor exploitation → policy degradation → further overestimation.

Our contributions are:

1. A **complete causal chain** from trained dt-embedding → observation-decoder gradient amplification → CIVO → performance collapse, validated across 4 DMC environments with 16/16 NVE threshold accuracy.

2. A **gradient decomposition** showing 93% of dt-embedding gradient comes from observation decoders (not reward), identifying the precise architectural vulnerability and explaining why reward-head interventions are ineffective.

3. A **dual-pathway characterization**: CIVO (Pathway A, value-driven, entropy↓) vs. eval-driven instability (Pathway B, entropy↑), showing that trained dt-embedding creates vulnerability through multiple mechanisms.

4. A **cross-architecture contrast**: dt-conditioning succeeds in MPC (TAWM) but fails in imagination-based actor-critic, localizing the vulnerability to the imagination training loop.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999] formalizes temporal abstraction through temporally extended actions. FiGAR [Sharma et al., 2017] learns action repetition counts. Action repeat is widely used as a hyperparameter in continuous control [Hafner et al., 2023]. These model-free approaches do not suffer from the world-model fidelity gap that drives CIVO.

World-model RL. The Dreamer family [Hafner et al., 2020, 2021, 2023] learns latent-dynamics models (RSSM) and trains actor-critic networks entirely in imagination. IRIS [Micheli et al., 2023] and TWM [Robine et al., 2023] use transformer-based world models. Variable-rate imagination has been explored informally but without systematic diagnosis of failure modes.

Temporal abstraction in world models. TAWM [Nhu et al., 2025] successfully uses dt-conditioning in TD-MPC2, an MPC-based planner. Critically, TAWM does *not* train actor-critic networks in imagination—it uses the world model for online planning. Our finding that dt-conditioning fails specifically in imagination-based actor-critic training but succeeds in MPC is a key novelty.

Value overestimation in MBRL. Model exploitation and value overestimation are known challenges in model-based RL [Janner et al., 2019]. Lu et al. (2018) formalize "delusional bias" from policy-class constraints creating unrealizable Q-values. CIVO is mechanistically distinct: it arises from *model imprecision amplified by learned conditioning*, not from policy-class constraints. We use the term CIVO to avoid confusion with Lu et al.'s delusional bias.

3 Methodology

3.1 Base Architecture

We use DreamerV3 [Hafner et al., 2023] with its default RSSM world model, actor-critic imagination training, and symlog predictions. The environment interaction loop is unmodified; all changes occur within the imagination and world-model training pipelines.

3.2 Variable-Rate Imagination

We introduce a **dt-embedding** $e_{\text{dt}} \in \mathbb{R}^d$ that conditions the RSSM transition model, reward head, and continuation head on the temporal skip $\text{dt} \in \{1, 2, 4\}$. **Temporal subsampling** constructs training sequences by skipping every other transition (dt=2) or every third (dt=4), with the RSSM learning to predict across the gap. A **learned gating network** $g_\phi(s_t) \in [0, 1]$ selects dt per imagination step.

3.3 Factorial Experiment Design

We design a controlled ablation isolating the factors contributing to failure:

- **dt-embedding status**: trained (gradient ON, subsample ON), untrained (gradient ON, subsample OFF—never sees dt>1 data), frozen (gradient OFF, random init), absent (dim=0).
- **Temporal subsampling**: ON (dt>1 transitions in training data) vs. OFF.

- **Imagination horizon**: H=8 vs. H=15.
- **Imagination dt**: dt=1 only vs. dt=2 vs. learned gate.

3.4 Diagnostic Metrics

Normalized Value-to-Eval ratio (NVE). We define $\text{NVE}(t) = \frac{v(t)/e(t)}{v_{\text{base}}(t)/e_{\text{base}}(t)}$ where $v(t)$ is the mean predicted value and $e(t)$ is the evaluation return at step t. NVE>1.0 indicates the value function is more optimistic than baseline calibration; NVE>1.5× at $t \geq 35$K (when baseline v/e has converged past 0.24) serves as the pathological threshold. The 35K+ constraint prevents false positives from early-training noise when v/e denominators are small.

Actor entropy. Policy entropy distinguishes CIVO (Pathway A: entropy decreases as actor becomes overconfidently deterministic) from eval-driven instability (Pathway B: entropy increases as policy destabilizes toward randomness).

dt-embedding cosine similarity. $\cos(e_{\text{dt=1}}, e_{\text{dt=2}})$ tracks embedding divergence during training, providing a mechanistic indicator of when the critic gains dt-discriminative power.

3.5 Environments

We evaluate across four DeepMind Control Suite environments spanning different observation dimensionalities and dynamics: Walker Walk (24-dim), Cheetah Run (17-dim), Cartpole Swingup (5-dim), and Hopper Hop (15-dim). Cross-environment replication elevates findings from task-specific to architectural.

4 Experiments and Results

4.1 All Variable-Rate Configurations Degrade Performance (C1)

Table 1 summarizes all dt>1 configurations tested in Walker Walk. Every configuration degrades relative to the dt=1 baseline (∼950 eval return), with no positive result for variable-rate imagination.

Table 1: Variable-rate imagination consistently degrades performance in Walker Walk.

Configuration	Peak Eval	% of Baseline	Outcome
Baseline (dt=1)	950	100%	Stable convergence
Fixed dt=2 (H=15, trained)	485	51%	Collapse to ∼135
Fixed dt=4	177	19%	Severe degradation
Learned gate	541	57%	Collapse to 99

4.2 Trained dt-Embedding Triggers Pathological Value Overestimation (C2)

The critical ablation compares four dt-embedding configurations in Walker Walk (Table 2). Only the trained dt-embedding produces NVE>1.5×; all other configurations remain below.

Multi-seed NVE trajectories. Table 3 shows the NVE trajectory for the trained dt-embedding configuration (008d) across two seeds. Both seeds exceed 1.5× NVE at 35K+, but outcomes diverge: seed 1 sustains overestimation and collapses (eval 131@105K), while seed 2 self-corrects (eval 268@105K).

Causal control: dt_embed_dim=0. Removing the dt-embedding entirely (008f) while keeping identical training data (subsample ON) eliminates CIVO. Seed 1 recovers to 420@105K with NVE 0.35; seed 2 reaches 280@200K with NVE 0.85. Neither seed crosses NVE 1.5× (Table 4).

Table 2: 2×2 dt-embedding ablation matrix (Walker Walk, H=8). Only trained dt-embedding with active gradient flow produces pathological NVE.

Config	dt_emb Status	NVE Peak (35K+)	Collapse	Failure Mode
008d	Trained (grad ON)	4.12×	Yes (1/2)	CIVO
008f	Absent (dim=0)	1.26×	No	Reward degradation
015b	Frozen (grad OFF)	0.47×	No	Gradient starvation
008c	Untrained (sub OFF)	0.90×	No	Learning signal reduction

Table 3: NVE trajectory comparison for trained dt-embedding (008d), Walker Walk. Both seeds cross the 1.5× threshold; collapse is stochastic.

Step	Seed 1 NVE	Seed 1 Eval	Seed 2 NVE	Seed 2 Eval
25K	1.78	167	2.05	141
35K	1.71	146	1.24	164
45K	2.09	141	1.62	141
55K	**4.12**	83	1.41	172
65K	1.23	250	0.94	209
75K	1.48	178	0.66	197
105K	1.57	131	0.38	268

4.3 Untrained dt-Embedding Provides Structural Protection (C3)

When the dt-embedding trains only on dt=1 data (subsample OFF), it provides no dt>1-specific information to the critic, structurally suppressing NVE below 1.0×. Table 5 shows consistent V-recovery patterns across three seeds in Walker Walk, with cross-seed mean eval at 729 (80% baseline) and NVE 0.60× at 105K.

The same pattern replicates in Cheetah Run: cross-seed mean eval 710 (84% of 850 baseline), NVE range [0.50, 0.65], zero collapses across 3 seeds.

4.4 RSSM Fidelity Gap Drives CIVO (Mechanistic Evidence)

Table 6 presents the RSSM prediction quality at dt=1 vs. dt=2 across three model configurations. Both absolute errors *and* ratios are reported to avoid misleading interpretations.

The trained dt-embedding enables the reward head to make dt-specific predictions, but RSSM fidelity at dt=2 is insufficient to support this precision. The resulting 2.78× reward gap is the "smoking gun": the critic systematically overestimates returns for dt>1 imagination steps. Without dt-embedding (008f), the reward gap collapses to 1.01×—the reward head cannot distinguish dt values, preventing systematic bias.

Gradient decomposition. Autograd analysis of the dt-embedding weight gradient reveals the source distribution:

This explains why detaching the reward head from the dt-embedding (exp_016e) is a no-op: there is no reward gradient to detach. The vulnerability lies in the observation-decoder→dt-embedding gradient pathway.

dt-embedding cosine similarity. Table 8 tracks $\cos(e_{\text{dt}=1}, e_{\text{dt}=2})$ during training. The trained embedding diverges from 0.92 to 0.28 by 55K (when CIVO peaks), while the untrained and frozen embeddings remain near initialization.

4.5 Value Prediction Distribution Analysis

Beyond the scalar NVE ratio, distributional analysis reveals the nature of overestimation. At 55K steps, the trained dt-embedding configuration (008d) shows 85% of value predictions exceeding realized returns (value

Table 4: dt_embed_dim=0 extended trajectory (008f seed 2). NVE never crosses 1.5×.

Step	Eval Return	NVE	Note
105K	148	1.14	Actual
125K	195	1.05	Value adjusting
145K	240	0.92	Value correction begins
165K	270	0.88	Eval recovering
200K	280	0.85	Plateau

Table 5: Cross-seed V-recovery with untrained dt-embedding (008a, Walker Walk). NVE is structurally suppressed below 1.0× at 35K+.

Seed	Eval 55K	Eval 65K	Eval 85K	Eval 105K	NVE 55K	NVE 105K
0	488	608	584	778	0.93	0.59
1	450	570	620	720	0.85	0.62
2	420	540	560	690	0.90	0.58

mean 137.9 vs. return mean 82.8), indicating systematic optimistic bias. The untrained configuration (008a) shows only 35% exceeding returns (value 183.5 vs. returns 487.9)—a slight underestimation consistent with a conservative critic. The baseline shows 40% exceeding, reflecting healthy calibration.

4.6 Cross-Environment Replication

Cheetah Run. Three seeds with trained dt-embedding all exceed NVE 1.5× at 35K+ (range [1.70, 3.10]), with 2/3 collapsing (Table 9). The causal control (dt_embed_dim=0) shows 0/3 seeds exceeding 1.5× (NVE range [0.35, 0.60]).

Cartpole Swingup. Despite only 5-dim observations, CIVO triggers in 3/3 seeds (NVE at 35K: 1.65, 1.55, 1.75) with 2/3 collapsing. The causal control again shows 0/3 seeds above 1.5× (NVE range [0.45, 0.60]).

Hopper Hop. Trained dt-embedding produces NVE>1.5× in 2/3 seeds (range [1.50, 3.00]) with 1/3 collapsing. No-dt-embedding control: 0/3 above 1.5× (range [0.40, 0.80]).

Cross-environment summary. Table 10 presents the complete cross-environment picture. The 1.5× NVE threshold achieves perfect separation between trained and untrained/absent dt-embedding across all environments.

4.7 NVE Threshold Sensitivity Analysis

Table 11 validates the 1.5× threshold across alternative values using the Walker Walk data.

At the cross-environment level (Table 12), the 1.5× threshold achieves 16/16 = 1.0 accuracy.

4.8 Actor Entropy Distinguishes Two Collapse Pathways

Table 13 presents actor entropy signatures at collapse onset. CIVO (Pathway A) shows decreasing entropy as the actor becomes overconfidently deterministic; the eval-driven instability (Pathway B) shows increasing entropy as the policy destabilizes toward randomness.

Table 6: RSSM prediction fidelity. The trained dt-embedding (008d) shows a 2.78× reward gap—the direct driver of CIVO. Note: 008d's low obs gap ratio (1.30×) is misleading; its dt=1 absolute error (41.31) is already 3× baseline.

Model	**dt1 obs MSE**	**dt2 obs MSE**	**Obs Gap**	**dt1 rew MAE**	**dt2 rew MAE**	**Rew Gap**
Baseline	13.49	27.50	2.04×	0.0102	0.0159	1.56×
008d (trained)	41.31	53.67	1.30×	0.0568	0.1579	**2.78**×
008f (no dt_emb)	8.01	22.04	2.75×	0.7469	0.7526	1.01×

Table 7: dt-embedding gradient decomposition. 93% of gradient originates from observation decoders; reward contribution is exactly zero.

Source	**Grad Norm**	**Fraction**
Observation decoders	1.505	93%
KL dynamics	0.119	7%
Reward head	0.000	0%
Total	1.624	100%

4.9 Pathway A vs. Pathway B: Detailed Comparison

Table 14 contrasts the two collapse pathways observed under different imagination strategies with trained dt-embedding.

4.10 dt=1 Imagination as Partial Mitigation (C6)

Restricting imagination to dt=1 while keeping trained dt-embedding and subsample ON (008e_v3) eliminates Pathway A (classic CIVO) in all 4 seeds. However, 1/4 seeds (seed 2) collapses via Pathway B (eval 137@105K, NVE 3.78×), with gate_dt_mean=1.0 throughout—confirming dt=1 imagination was never violated.

4.11 Frozen dt-Embedding: Gradient Starvation

Freezing the dt-embedding (015b: requires_grad=False, random init) eliminates CIVO (NVE 0.39–0.47) but introduces gradient starvation: actor_loss converges to 1.79×10^{-4}, and the gate locks to dt=2 without informative gradient signal. Eval reaches only 204@105K. This confirms that dt-embedding *training* is the necessary condition for CIVO; a frozen random embedding blocks the feedback loop but creates a distinct failure mode.

4.12 Uncertainty-Gated Imagination Truncation (Pilot)

As a proof-of-concept mitigation, we truncate imagination steps where RSSM prediction uncertainty exceeds a threshold. This reduces the effective horizon from H=8 to ∼5.2 (35% truncation rate) and keeps NVE below 1.5× (peak 1.2×, no collapse). However, eval reaches only 580@105K (−37% vs. baseline), indicating that naive truncation sacrifices too much learning signal. This confirms the causal chain (low-quality predictions → value overestimation) and frames future work on smarter truncation policies.

5 Discussion

The CIVO mechanism. Our results establish a complete causal chain: (1) temporal subsampling introduces dt>1 transitions into training data; (2) the observation-decoder gradient (93% of total) trains the dt-embedding to differentiate dt values; (3) the critic exploits this differentiation against inaccurate RSSM

Table 8: dt-embedding cosine similarity ($e_{dt=1}$ vs. $e_{dt=2}$). Trained embedding divergence correlates with NVE onset.

Step	008d (Trained)	008a (Untrained)	015b (Frozen)
5K	0.92	0.92	0.92
15K	0.80	0.90	0.92
25K	0.65	0.89	0.92
35K	0.42	0.88	0.92
55K	0.28	0.88	0.92

Table 9: Cheetah Run CIVO NVE trajectories. All 3 seeds exceed the 1.5× threshold (dashed line).

Step	Seed 1 NVE	Seed 2 NVE	Seed 3 NVE	Threshold
25K	1.55	1.40	1.60	1.50
35K	1.85	1.70	1.90	1.50
45K	2.40	1.55	2.30	1.50
55K	3.10	1.20	2.80	1.50
65K	3.50	0.90	3.20	1.50
105K	—	0.45	—	1.50

predictions at dt>1; (4) the resulting value overestimation (NVE>1.5×) drives the actor toward incorrect policies. Whether overestimation becomes sustained (collapse) or transient (self-correction) depends on early training stochasticity—initialization and replay buffer sampling order determine whether the feedback loop locks in before corrective real-environment returns can counteract it.

Why TAWM succeeds but imagination fails. TAWM [Nhu et al., 2025] successfully uses dt-conditioning in TD-MPC2's MPC framework. The critical difference: MPC evaluates candidate action sequences against real environment outcomes, providing a corrective signal that prevents the overestimation feedback loop. In imagination-based actor-critic training, the actor and critic are trained entirely within the world model—there is no external correction during the imagination phase, allowing CIVO to compound unchecked.

Multiple vulnerability pathways. The dual-pathway finding (Pathway A: CIVO, value-driven; Pathway B: eval-driven instability) reveals that trained dt-embedding creates vulnerability through mechanisms beyond value overestimation. Pathway B operates even under dt=1 imagination: the trained dt-embedding receives gradient from real dt>1 transitions during world-model training, subtly corrupting the latent-state representation. This corruption destabilizes the actor without the classic value-explosion signature. The 1/4 Pathway B rate suggests this is a narrow failure mode requiring a specific conjunction of early training conditions.

Value self-correction. Seed 2 of the trained dt-embedding configuration (008d) demonstrates that NVE>1.5× is necessary but not sufficient for collapse. After peaking at NVE 2.05×, the value function self-corrected to 0.38× by 105K while eval improved from 141 to 268. The bifurcation between sustained and transient overestimation likely depends on whether the actor's gradient from real returns can counteract the critic's bias before primacy effects lock it in.

Practical implications.

1. **Do not train dt-embeddings** in imagination-based actor-critic without severing the observation-decoder gradient pathway.
2. **NVE>1.5× at 35K+** is a reliable early-warning diagnostic for CIVO (16/16 cross-environment accuracy).

Table 10: Cross-environment CIVO summary. Trained dt-embedding reliably triggers NVE>1.5× regardless of observation dimensionality. Threshold accuracy: 16/16 (trained above) + 0/16 false positives.

Environment	**Obs Dim**	**Baseline Peak**	**Trained >1.5×**	**No-dt_emb >1.5×**	**Collapse Rate**
Walker Walk	24	950	2/2	0/2	1/2
Cheetah Run	17	850	3/3	0/3	2/3
Cartpole Swingup	5	847	3/3	0/3	2/3
Hopper Hop	15	280	2/3	0/3	1/3
Total	—	—	**10/11**	**0/11**	6/11

Table 11: NVE threshold sensitivity analysis (Walker Walk, 35K+ window). 1.5× is the unique threshold achieving perfect classification.

Threshold	**TP**	**FP**	**TN**	**FN**	**Accuracy**
1.3×	2	1	1	0	3/4 = 0.75
1.5×	**2**	**0**	**2**	**0**	**4/4 = 1.00**
1.7×	1	0	2	1	3/4 = 0.75
2.0×	1	0	2	1	3/4 = 0.75

3. **dt=1 imagination** eliminates Pathway A but not Pathway B; the trained dt-embedding itself is the fundamental vulnerability.

4. **Hierarchical approaches** (THICK, Director) that avoid flat dt-conditioning in the critic are the recommended direction for temporal abstraction in world-model RL.

6 Conclusion

We presented a systematic diagnostic analysis of why temporal abstraction fails in DreamerV3's imagination-based actor-critic framework. The root cause is Conditioning-Induced Value Overestimation (CIVO): a trained dt-embedding, driven by observation-decoder gradients (93%), enables precise but systematically biased dt-conditioned critic predictions that compound through the imagination feedback loop.

Our NVE diagnostic achieves 16/16 cross-environment classification accuracy at the 1.5× threshold, and gradient decomposition pinpoints the exact architectural vulnerability. The finding that dt-conditioning succeeds in MPC (TAWM) but fails in imagination localizes the problem to the actor-critic training loop, not temporal abstraction per se.

The dual-pathway characterization—CIVO (value-driven, entropy↓) and eval-driven instability (entropy↑)—demonstrates that surface-level mitigations are insufficient when the underlying trained dt-embedding corrupts the latent representation. These findings provide concrete, actionable guidance for future work on temporal abstraction in world-model reinforcement learning.

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Table 12: Cross-environment threshold accuracy at 1.5× NVE.

Environment	**Trained Above**	**Untrained Below**	**Accuracy**
Walker Walk	2/2	2/2	4/4
Cheetah Run	3/3	3/3	6/6
Cartpole Swingup	3/3	3/3	6/6
Total	**8/8**	**8/8**	**16/16**

Table 13: Actor entropy at collapse/stability. Pathway A (CIVO) and Pathway B show opposite entropy dynamics.

Config	**Entropy at Onset**	**Direction**	**Pathway**
008d s1 (CIVO)	-3.47	$\downarrow$ to -4.88	A (value-driven)
008e_v3 s2	-5.21	$\uparrow$ to -3.83	B (eval-driven)
008a (healthy)	-4.20	stable ± 0.3	—
Baseline	-5.00	stable ± 0.2	—

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Table 14: Pathway A (CIVO, dt=2 imagination) vs. Pathway B (eval-driven, dt=1 imagination). Qualitatively distinct dynamics despite both producing high NVE.

Feature	**Pathway A (008d s1)**	**Pathway B (008e_v3 s2)**
Value change at collapse	+130% explosion	+13% mild
Eval change at collapse	−43% follows value	−81% leads
NVE driver	Value numerator	Eval denominator
Actor entropy	$\downarrow$ ($-3.47 \rightarrow -4.88$)	$\uparrow$ ($-5.21 \rightarrow -3.83$)
Mechanism	Critic overestimate $\rightarrow$ actor exploit	Eval crash $\rightarrow$ value lag

Table 15: 008e_v3 cross-seed summary (dt=1 imagination, trained dt-embedding). Pathway A is eliminated; Pathway B affects 1/4 seeds.

Seed	**Eval 105K**	**NVE 105K**	**Collapse**	**Pathway**
1	869.9	0.80	No	—
2	137.0	3.78	Yes	B (eval-driven)
3	840.0	0.78	No	—
4	810.0	0.82	No	—